WebTestPilot: Agentic E2E Web Testing against Natural Language Specification by Inferring Oracles with Symbolized GUI Elements

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Paper version: arXiv 2602.11724v3 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *it is hard to distinguish whether it stems from the hallucination or a real application bug*

An LLM tester must be **its own oracle**, but stochastic verdicts flake: five LLMs, ten trials each, gave seven distinct answers on one page. Prior testers never assert, or keep unstructured memory.

Motivation: two oracle gaps

1. **No oracle.** NaviQAte, LaVague drive; never verify.
2. **Flat memory.** PinATA: unstructured, capped.
3. **Flaky verdicts.** Same prompt, new answers.

Evaluation setup

4 live web apps · 100 NL requirements · 100 injected bugs · GPT-4.1 base · UI-Venus-7B grounding

- **Online:** live apps; steps auto-checked vs Playwright assertions.
- **Stop rule:** ends at scripted step count; sequential.

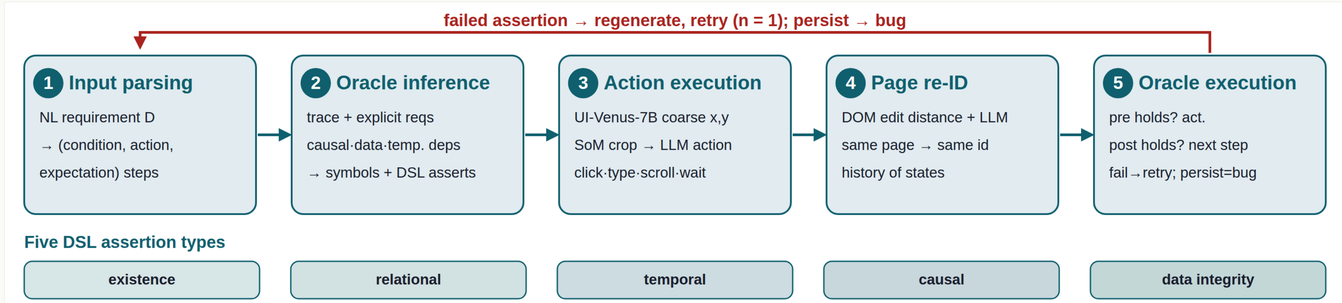
Our read (CUA Radar)

61	★★	online
score of 100	band	evidence level

- **Reports:** online eval, live apps; PinATA SOTA; cost; assertions checked.
- **Not reported:** held-out apps; seeds; ablation. AP8 = no.
- 2602.11724v3 · C18 · authors' numbers.

Method: a neurosymbolic oracle loop

- **Sees / acts:** full-page screenshot + DOM; click, type, press, scroll, wait on element ids.
- **Symbols:** Pydantic schemas abstract GUI elements, extracted on demand from past states.
- **Asserts:** Python-extended DSL pre/post per step; fail → retry (n = 1) → bug.



Our redraw (not the paper's figure): per-step loop; five assertion types in the DSL.

$$(s \models \mathrm{Pre} \wedge s \xrightarrow{a} s') \implies s' \models \mathrm{Post}$$

step success = precondition holds on s, postcondition on s'

Task completion (TC)

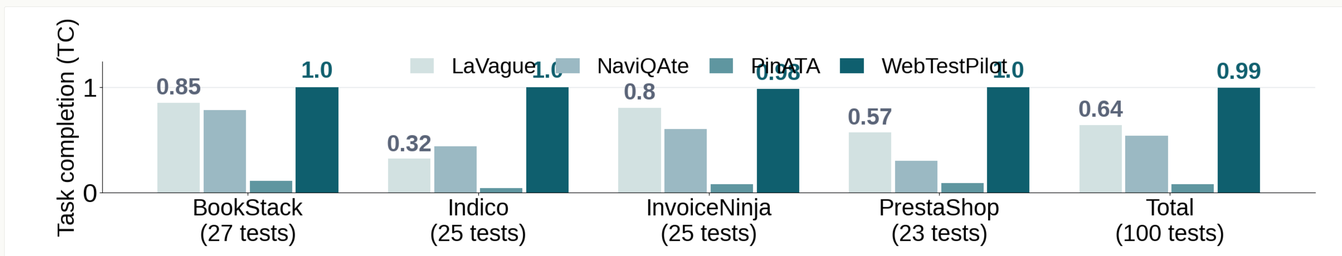


Table 3, redrawn.

+54.7% TC, +28.6% CT vs best baseline.

SO WHAT → *Correct-trace totals: 0.99 vs 0.77 (LaVague), 0.70 (NaviQAte).*

Headline Numbers

0.96
bug precision & recall
vs 0.26 / 0.69 (PinATA)

99% task completion
+70 precision gain
+27 recall gain

The paper's own controls

CONTROL	WEBTESTPILOT	COMPARED WITH
TC / CT totals	0.99 / 0.99	LaVague 0.64 / 0.77
Precision / recall	0.96 / 0.96	PinATA 0.26 / 0.69
23 replicated bugs	22 detected	PinATA 15
Median time / tokens	29 s / 10k	NaviQAte 40 s / 9k

Takeaway

Symbols + DSL pre/post assertions make the LLM its own oracle: 0.99 TC, 0.96 / 0.96 precision / recall.

assertions hand-checked · 23 real bugs replicated · robust 3B–72B

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